

Project Digest

Top 5 AI/ML-style prototypes for business impact

This digest is for leaders and peers who want proof before a long programme: five prototype patterns, each tied to a clear business outcome. Flagship builds on teoyongsong.github.io show how ideas move from sprint to something you can click, test, or pilot.

How flagship demos map to outcomes

Use this summary to brief non-technical stakeholders: what to expect and who it helps.

RAG Studio

Mechanism: Private Q&A on your own documents without sending files to the public cloud by default.

Outcome: Trust, speed of answers, auditability for sensitive content.

Stakeholders: Legal, HR, risk, IP-heavy teams; internal knowledge bases.

HDB resale price predictor

Mechanism: Indicative resale estimates from public transaction patterns and flat attributes.

Outcome: Faster, evidence-backed conversations on price expectations.

Stakeholders: Buyers, sellers, agents, policy or research discussions.

CNN waste classification

Mechanism: Image upload to a trained model for category-style feedback.

Outcome: Operational sorting support and education; visible "so what" from CV.

Stakeholders: Facilities, sustainability, training, compliance storytelling.

Online retail segmentation

Mechanism: RFM-style clusters with suggested actions on a large retail-style dataset.

Outcome: Marketing and ops alignment around segments, not averages.

Stakeholders: CRM, growth, analytics teams explaining "who" to target.

Once Upon (storytelling MVP)

Mechanism: Structured, personalized stories from child profile inputs.

Outcome: Engagement and product storytelling beyond charts; family-facing UX.

Stakeholders: Product, education, innovation demos to non-technical audiences.

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Top 5 AI/ML-style prototypes for business impact

Top 5 AI/ML-style prototypes (detail)

1. RAG Studio (private-by-default)

Problem: Teams need answers from internal documents without defaulting to public SaaS or leaking data.

Outcome / demo: A working pattern for retrieval-augmented Q&A with local-first posture; good for pilots.

Business impact: Reduces "we cannot use AI" friction in regulated or cautious environments.

Try it: <https://rag-model.streamlit.app/>

2. HDB resale price predictor

Problem: Resale discussions often rely on anecdotes; stakeholders want a repeatable, transparent baseline.

Outcome / demo: Supervised models on open data packaged for indicative estimates in a browser demo.

Business impact: Supports defensible conversations, not a substitute for professional valuation.

Try it: <https://hdb-prices-predictor.streamlit.app/>

3. CNN waste classification

Problem: Waste and recycling narratives need a tangible demo, not only slides.

Outcome / demo: Upload-to-prediction loop with a trained CNN for category-style feedback.

Business impact: Makes computer vision legible to ops and sustainability stakeholders.

Try it: <https://cnn-waste.streamlit.app/>

4. Online retail customer segmentation

Problem: One average customer hides segments that need different actions.

Outcome / demo: Unsupervised clustering (e.g. RFM) with profiles and suggested CRM-style moves.

Business impact: Bridges analytics and commercial decisions in a single narrative.

Try it: See [project page](#) for the hosted app link (portfolio: [customer segmentation](#)).

5. Once Upon (storytelling MVP)

Problem: Not every initiative is a dashboard; some wins are experiential and narrative.

Outcome / demo: Personalized story flow with profiles and structured sections in a Streamlit MVP.

Business impact: Shows product thinking and storytelling as a complement to ML and data work.

Try it: <https://story-telling.streamlit.app/>

More on the site

Full write-ups: teoyongsong.github.io. Speaking and consulting: dannyyteo.github.io. If you received this PDF as a welcome gift, new project notices arrive separately when work is added to the portfolio.